Università di Genova

Embedded Systems

Tools

Prof. Enrico Simetti

Tools

MPLAB X IDE: http://www.microchip.com/mplab/mplab-x-ide

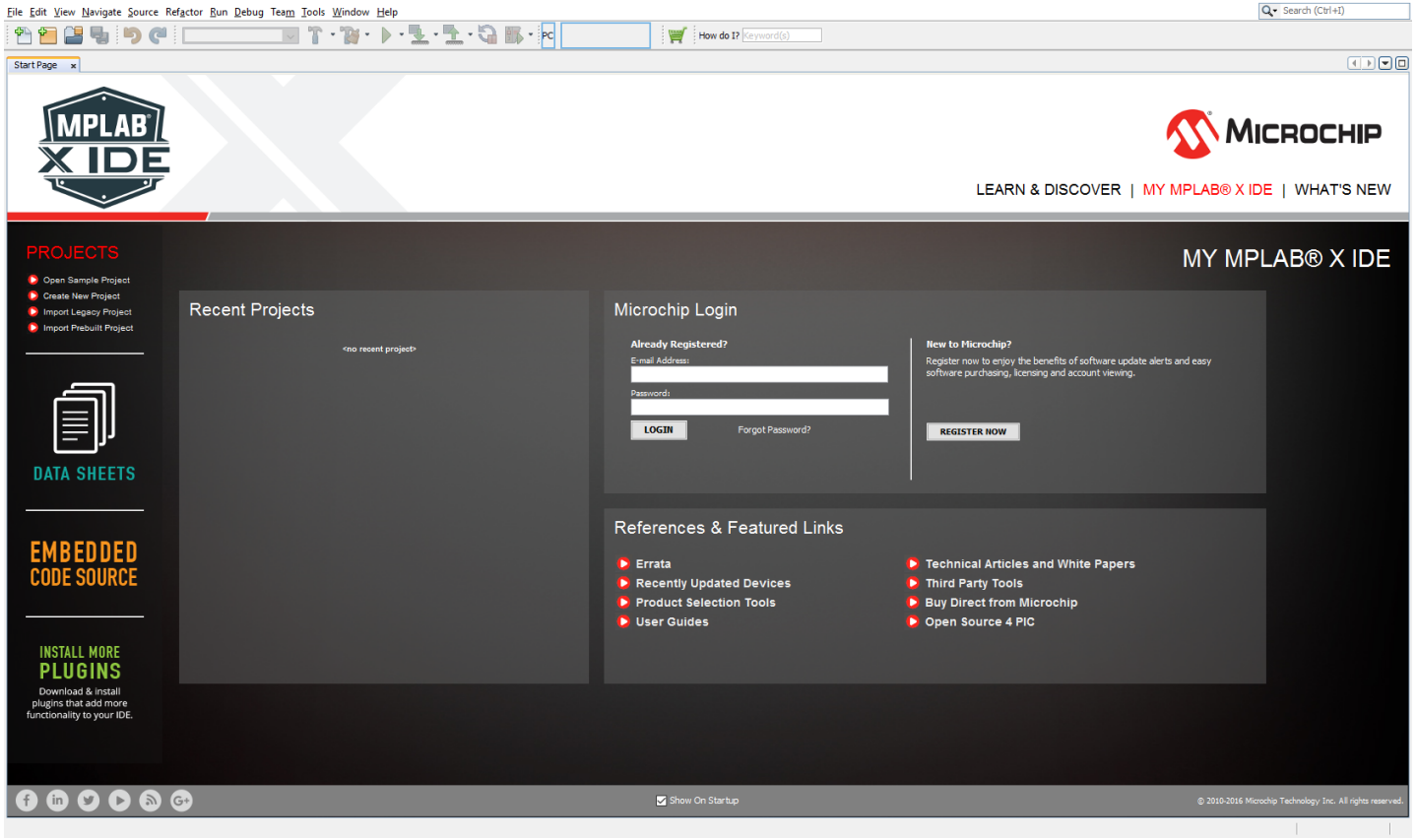
XC16 Compiler: http://www.microchip.com/mplab/compilers

MikroE mikroBootloader: https://www.mikroe.com/mikrobootloader

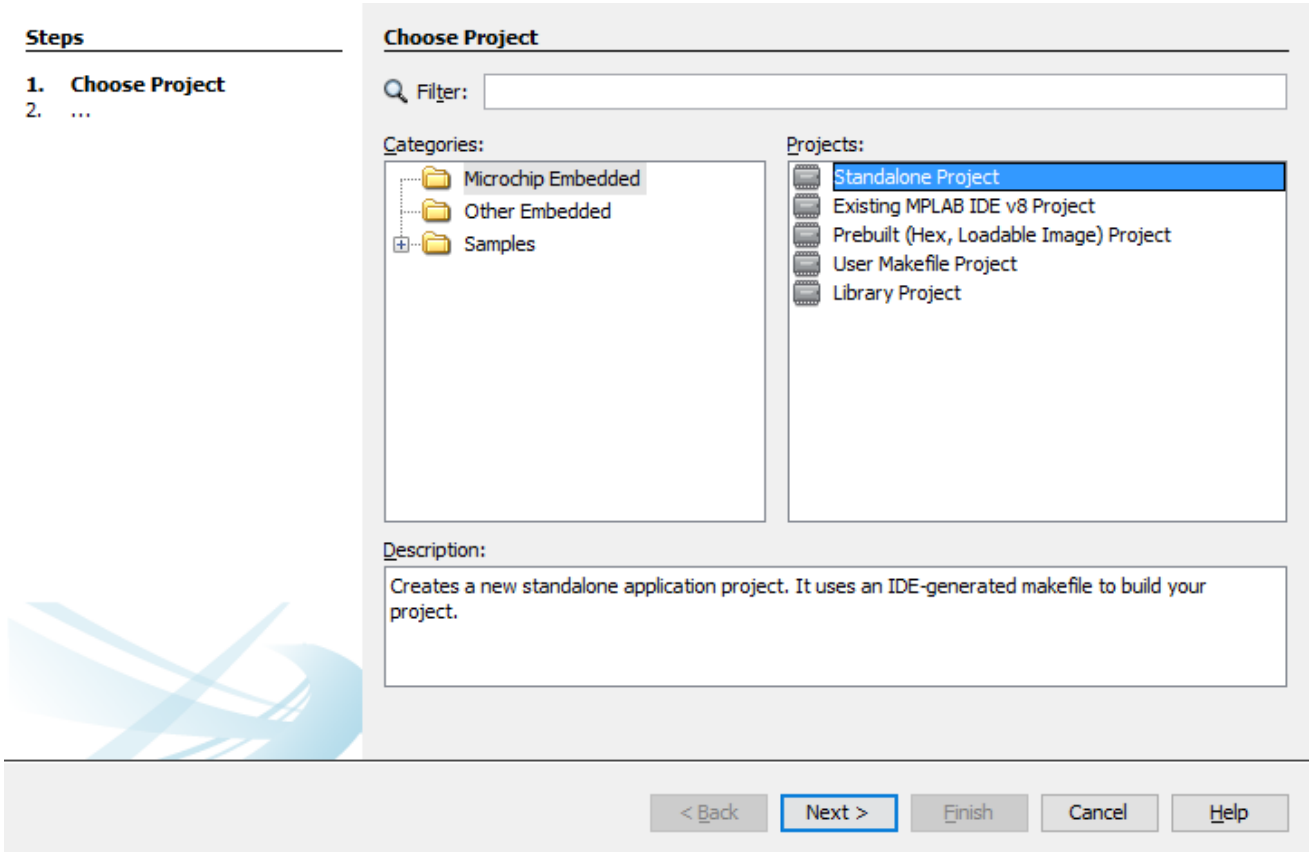
Now available: MPLAB for VS Code:

https://www.microchip.com/en-us/tools-resources/develop/mplab-tools-vs-code

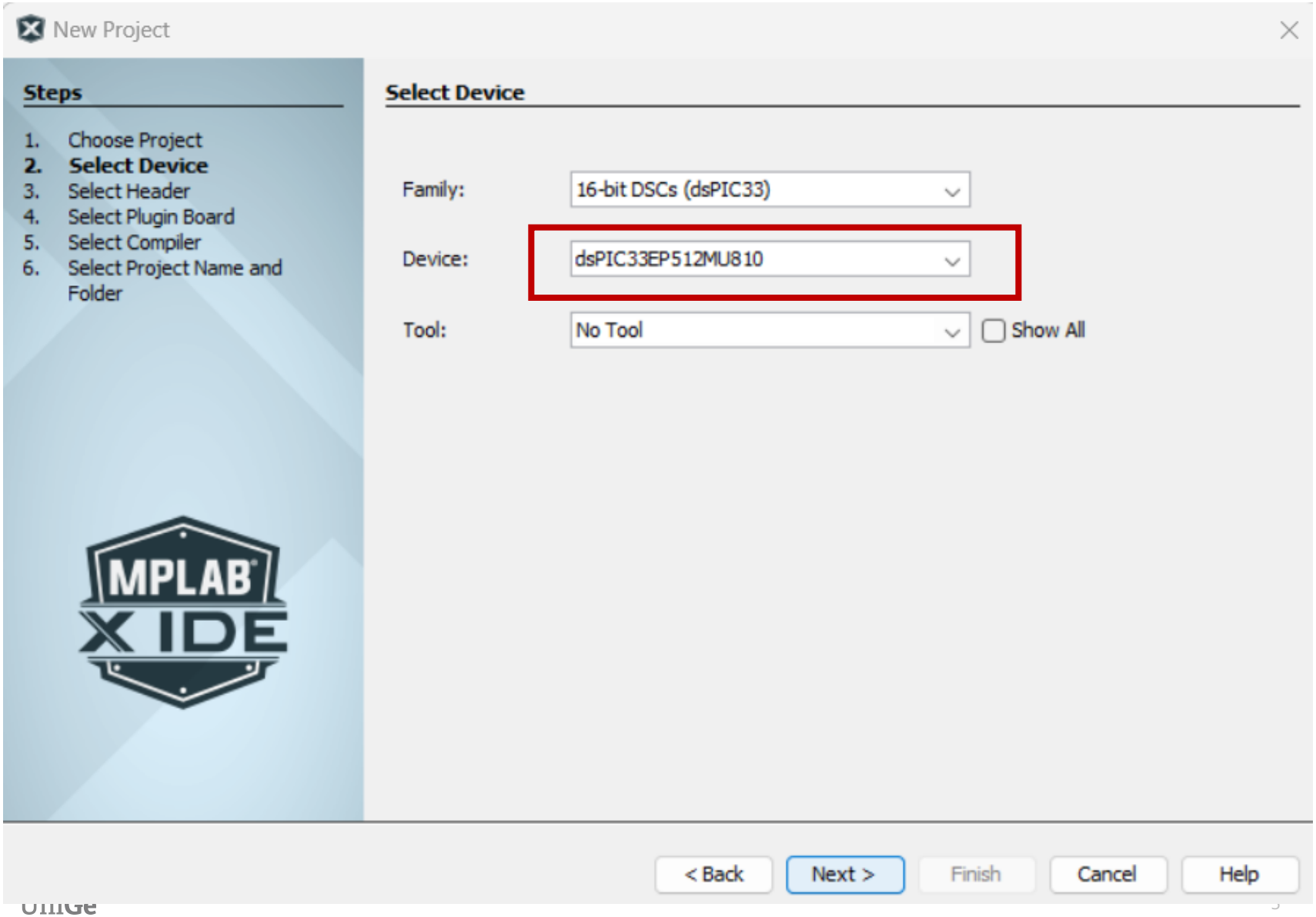
MPLAB X


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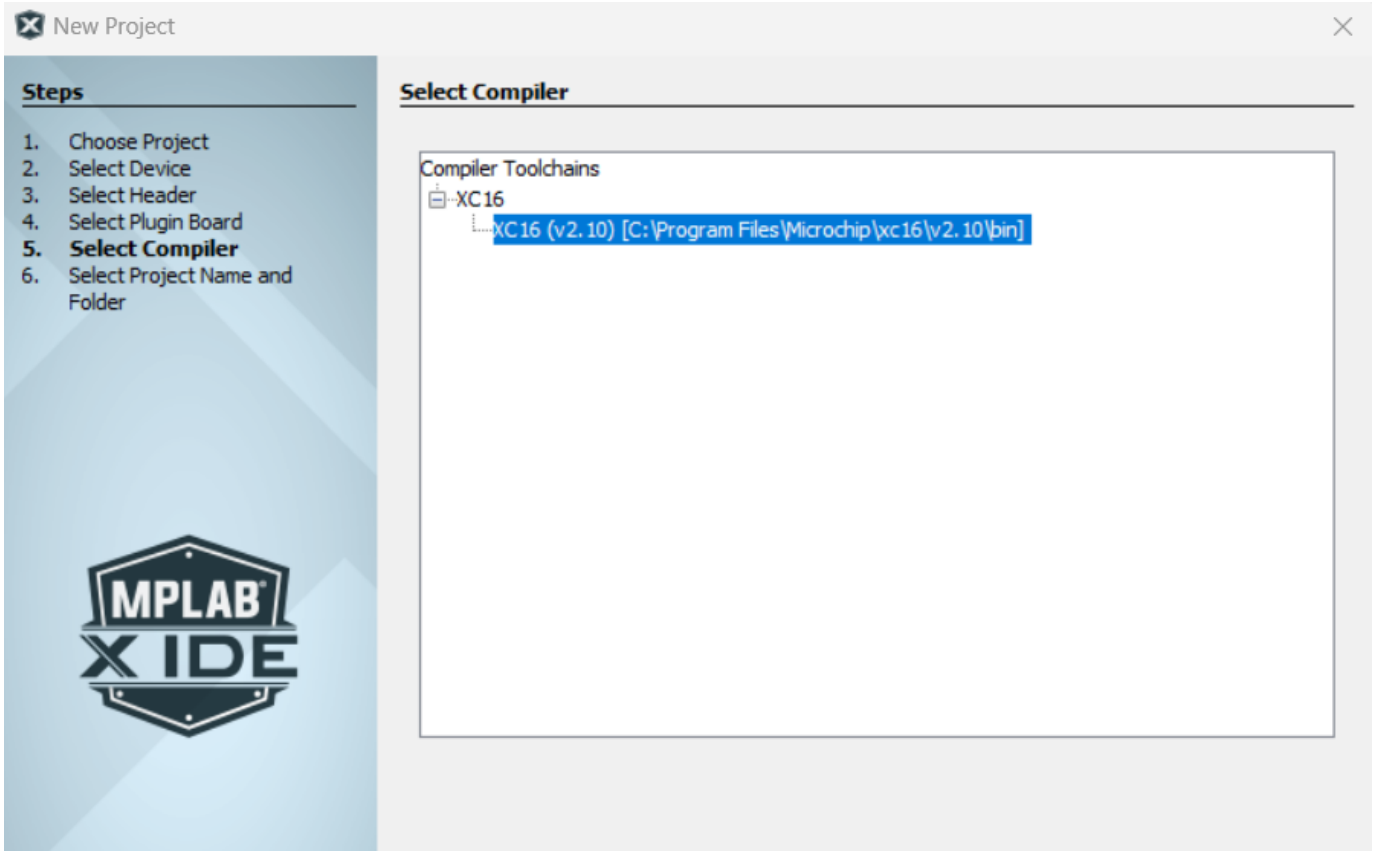
MPLAB X: Create a New Project



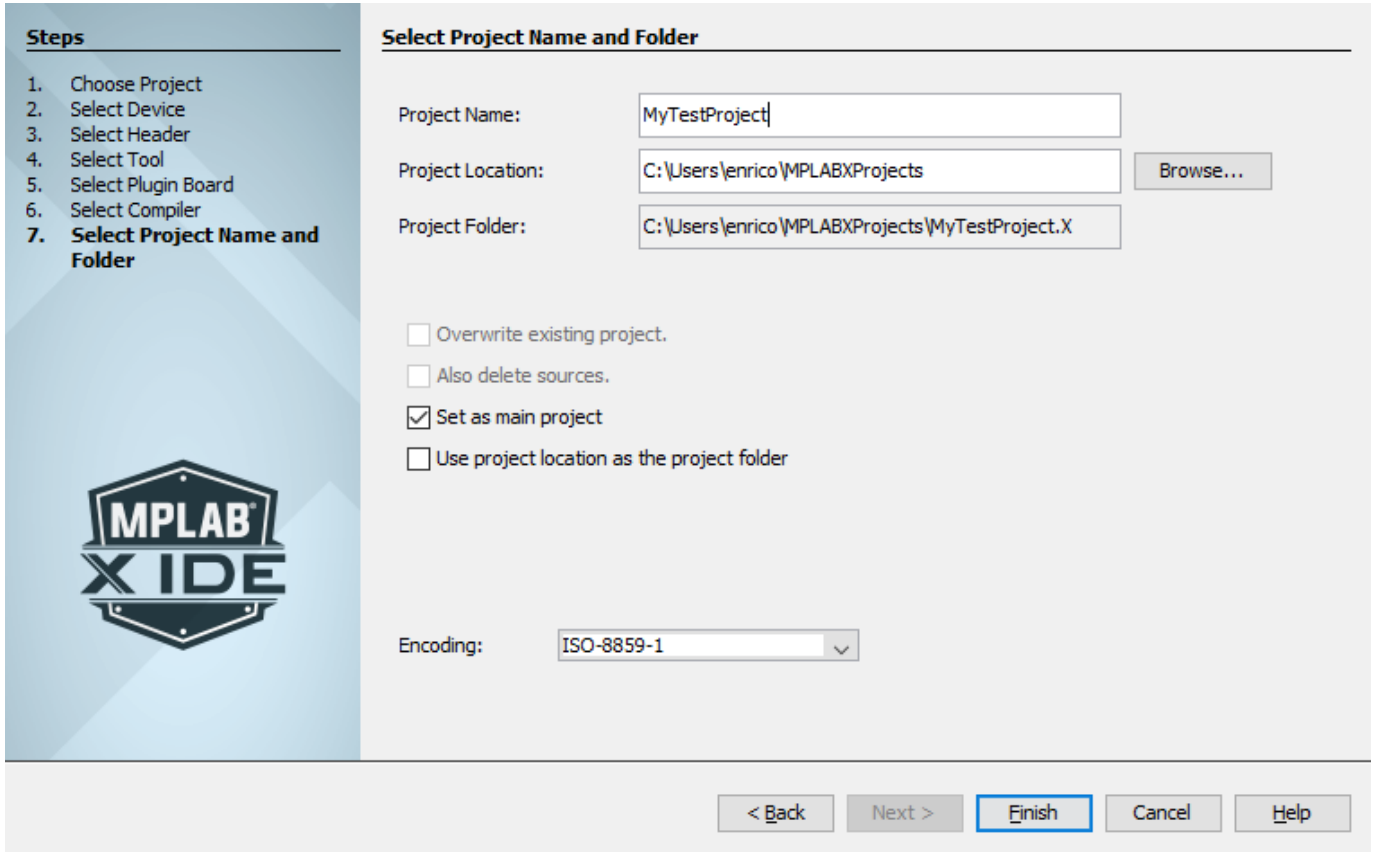
MPLAB X: Create a New Project



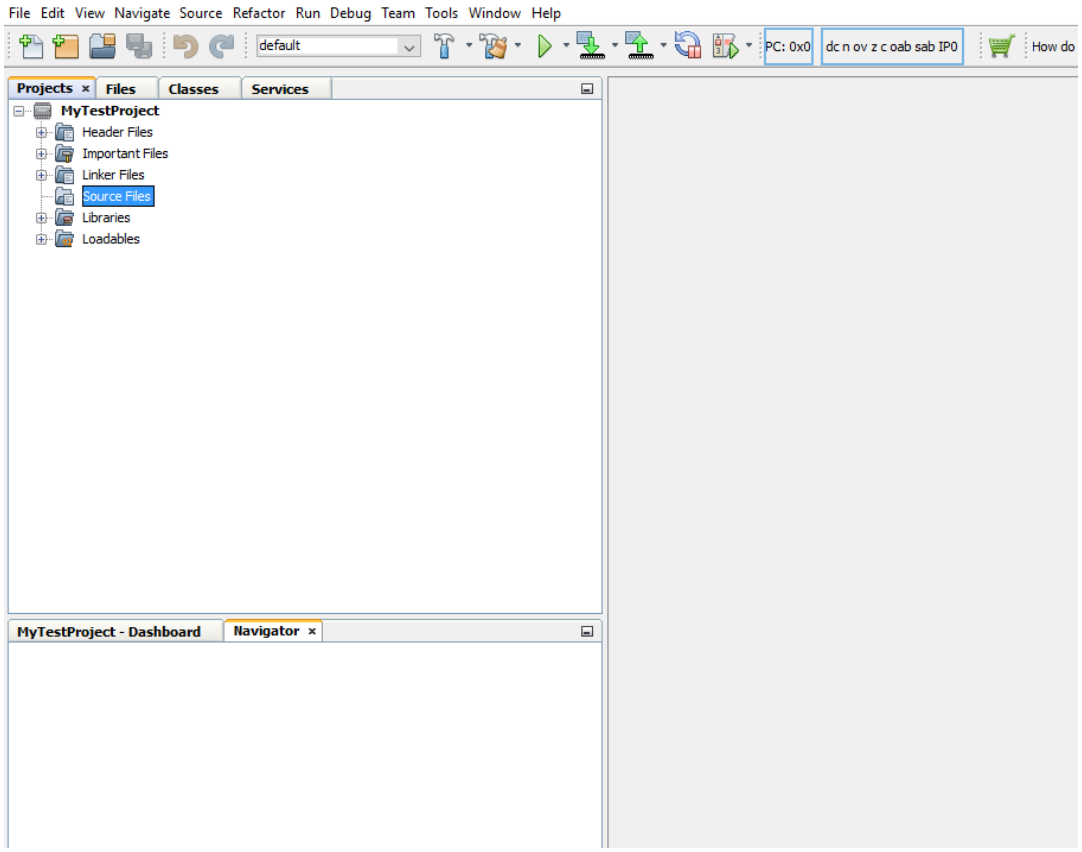
MPLAB X: Create a New Project



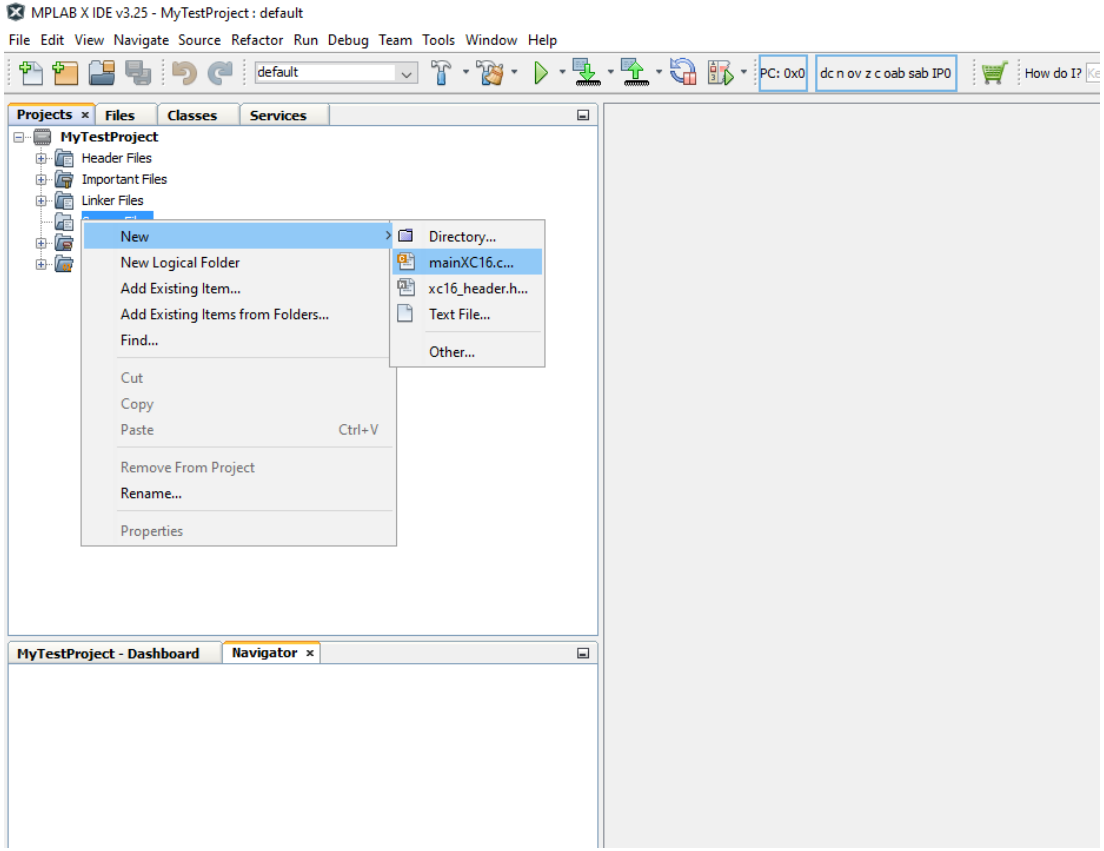
MPLAB X: Create a New Project



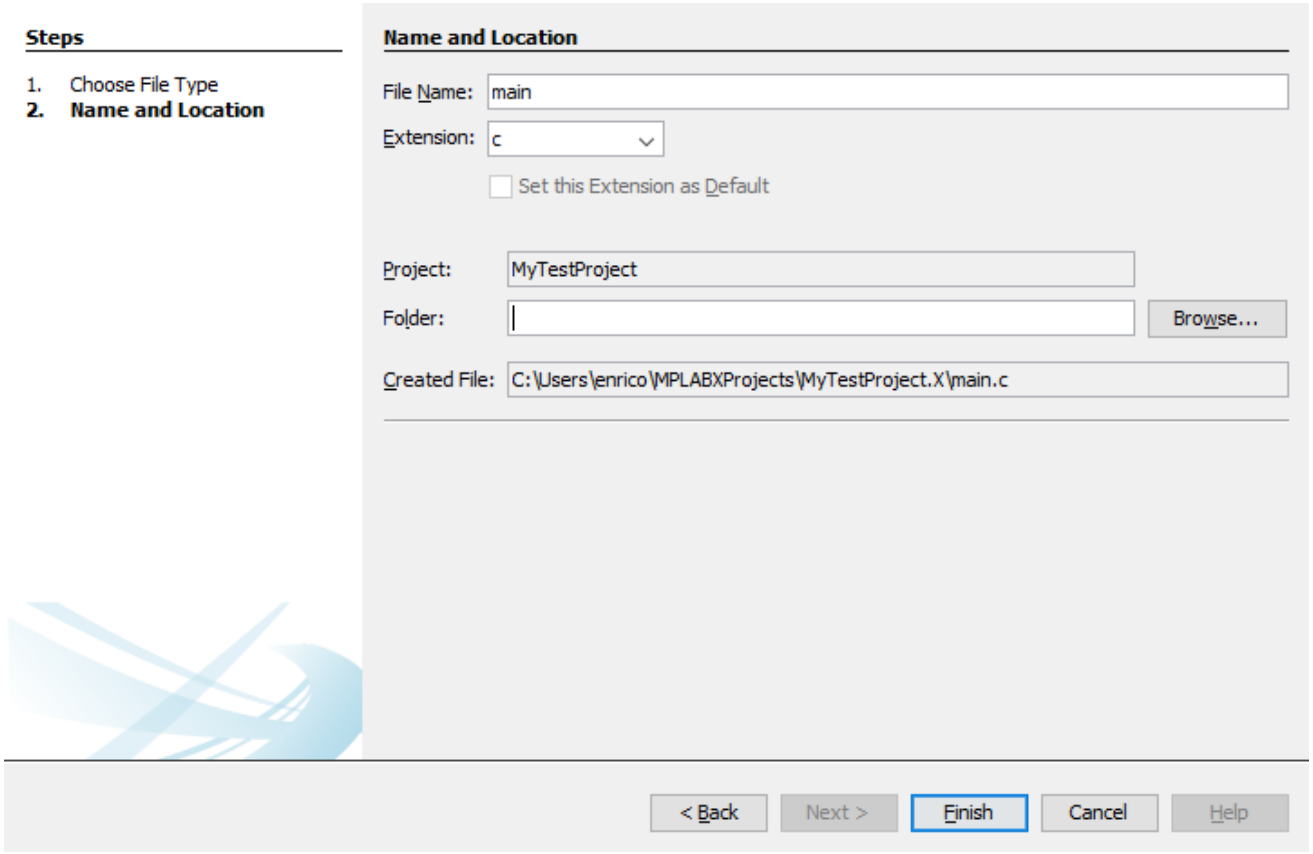
MPLAB X: Create a New Project



MPLAB X: Add a Source File

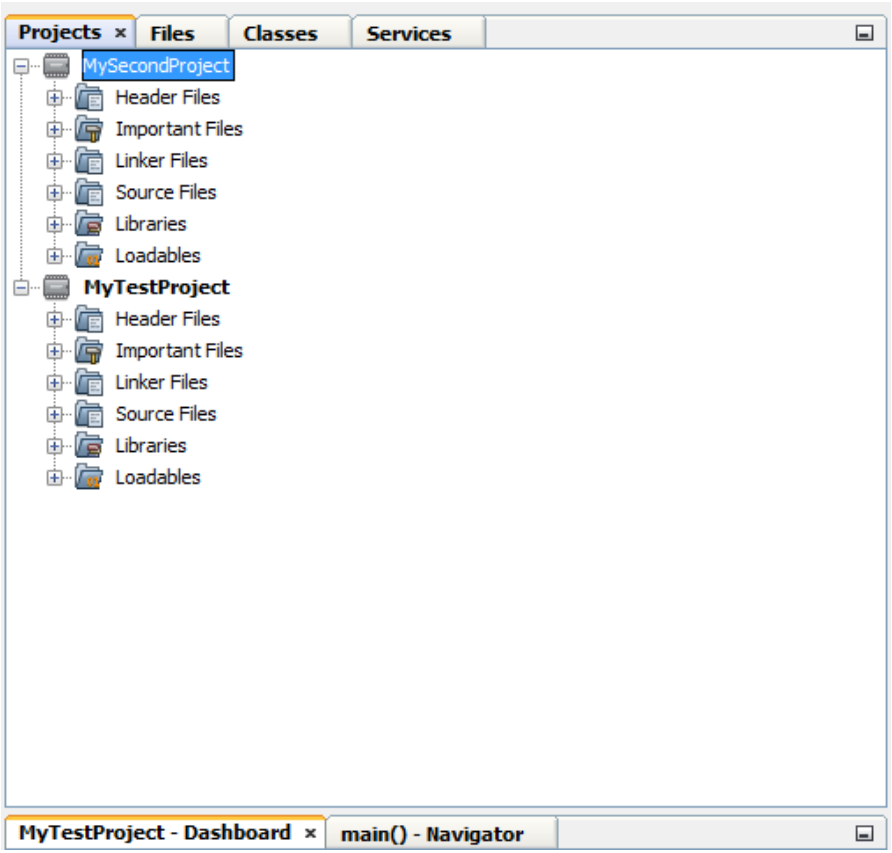


MPLAB X: Add a Source File



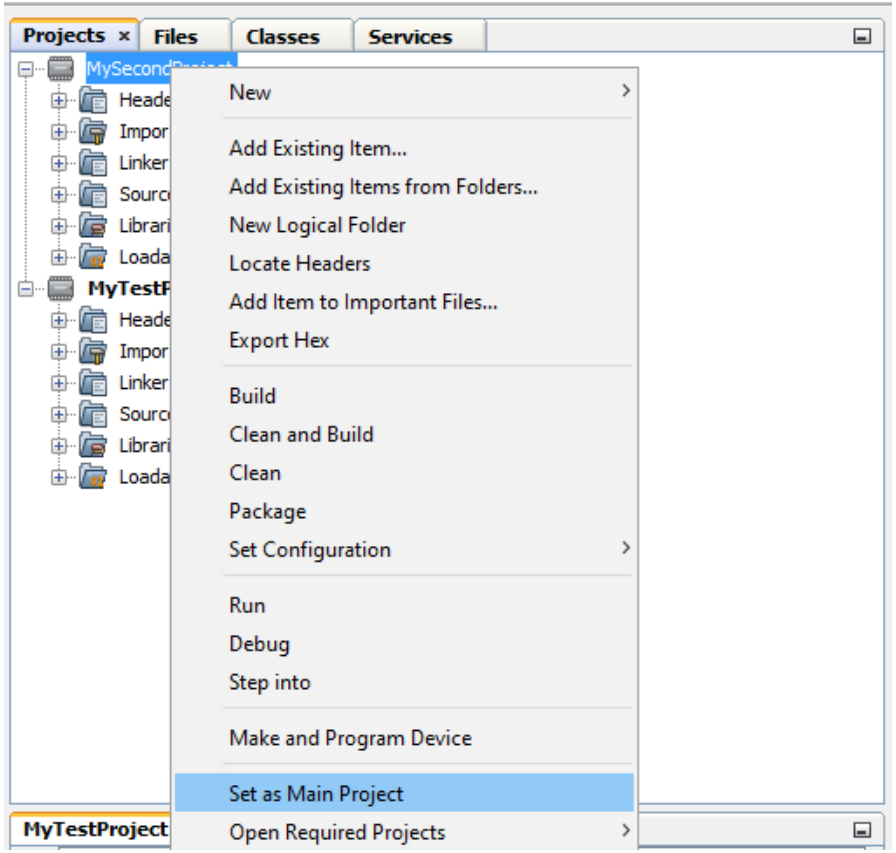
MPLAB X: Multiple Projects Management

Now the IDE is set to compile “MyTestProject”



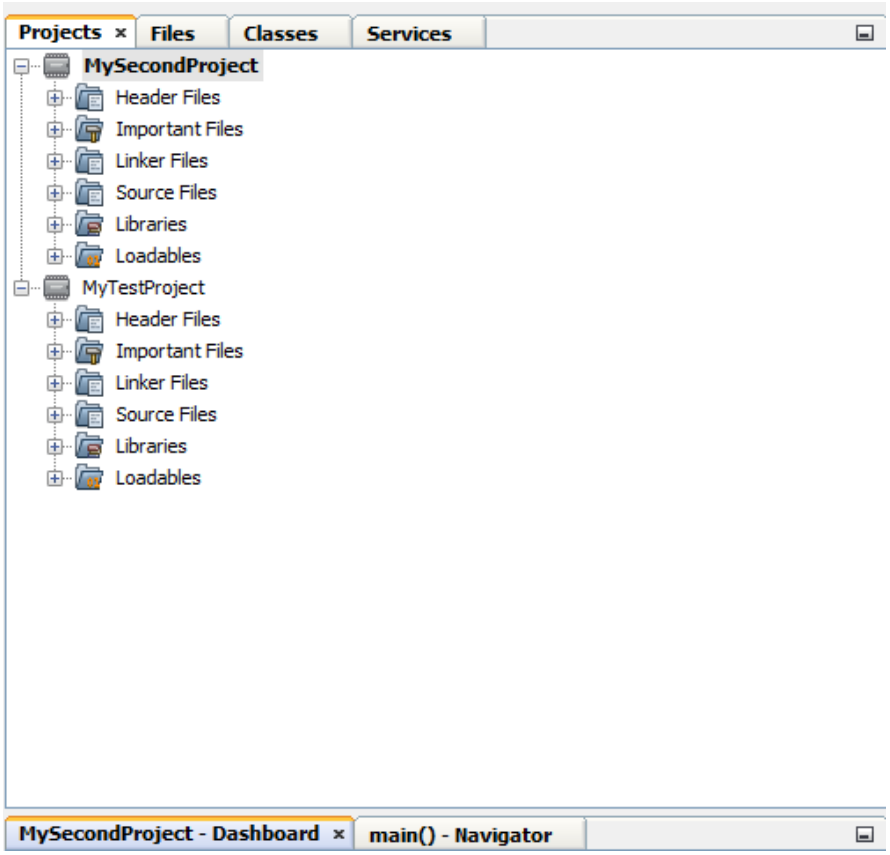
MPLAB X: Multiple Projects Management

If I want to compile “MySecondProject”’, right click and select "Set as Main Project"

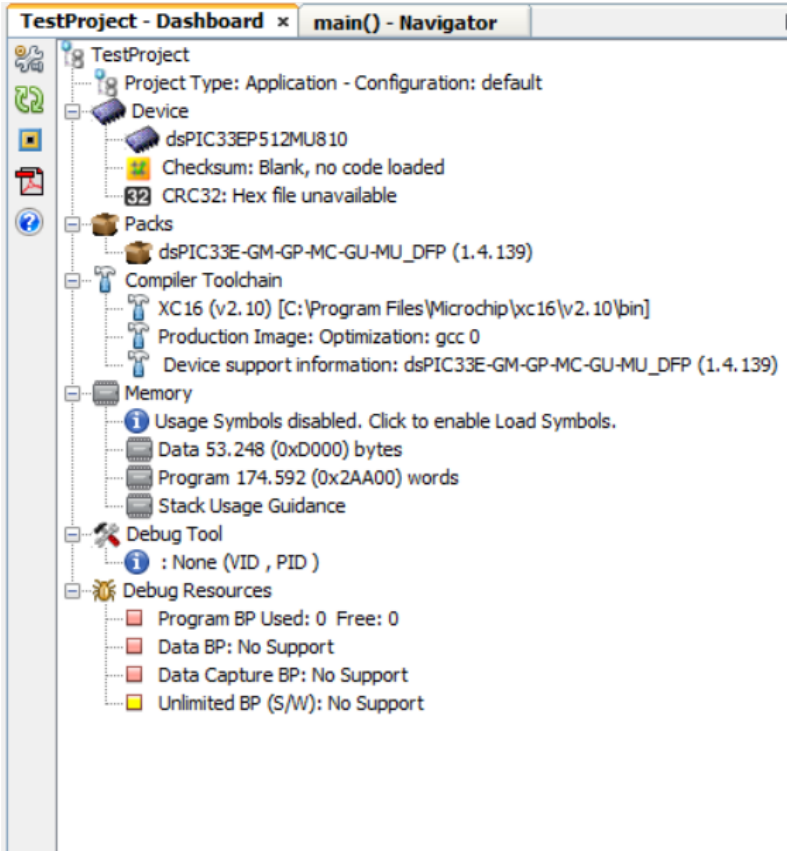


MPLAB X: Multiple Projects Management

Now “**MySecondProject”** is written in bold indicating that it is the main project



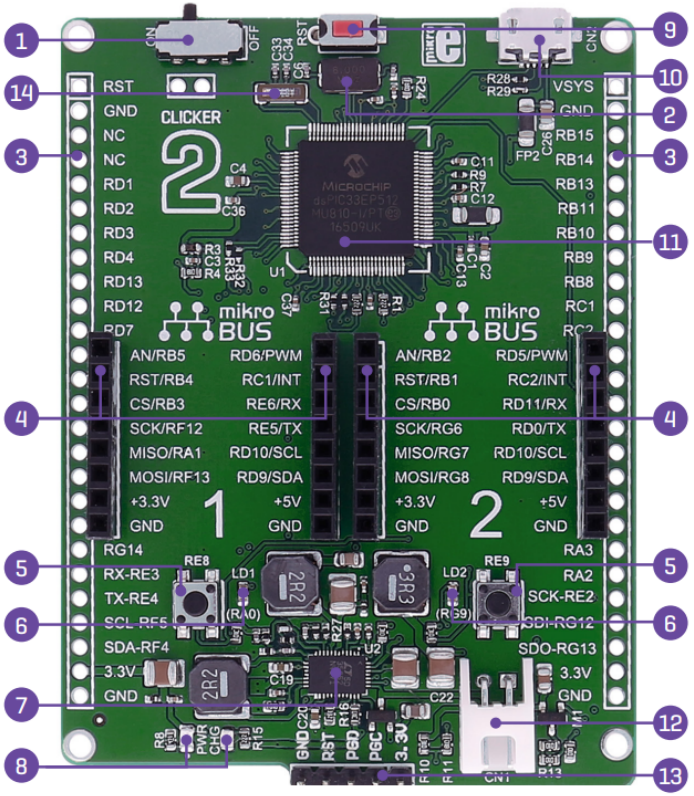
MPLAB X: Dashboard



Clicker2 Board

Key features

1. ON/OFF switch
2. 8 MHz crystal oscillator
3. two 1x26 connection pads
4. mikroBUS™ sockets 1 and 2
5. Pushbuttons
6. Additional LEDs
7. LTC3586 USB power manager IC
8. Power and Charge indication LEDs
9. RESET button
10. Micro USB connector
11. dsPIC33EP512MU810 MCU
12. Li-Polymer battery connector
13. mikroProg programmer connector
14. 32.768 KHz crystal oscillator



mikroBootloader

The Clicker2 board contains a **bootloader**, a light program stored permanently in memory that allows to flash firmware without a specific programmer. The mikroE mikroBootloader software is required to download the firmware on the board. It's only available on Windows platforms.



Upload your Firmware (Windows)

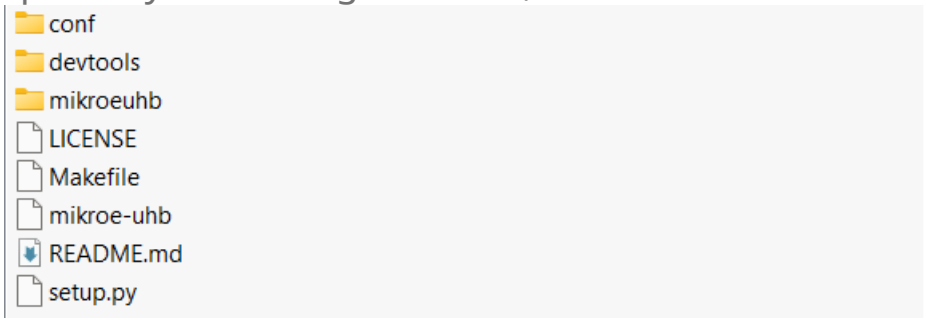
To write your program on the board use the following steps:

- Connect the board to your PC with the microUSB cable
- Press the RESET button (the red one) on the board
- The icon besides *Wait for the USB link* should become red. You have approx. 5 seconds to press the connect button on the interface. If you don't press it within 5 seconds, the board will run the previously flashed firmware
- Once you're connected, select *Browse for HEX* and look for the following path:
- C:/Users/yourusername/MPLABXProjects/projectname.X/dist/default/production/projectname.X.production.hex. You generate this file every time you build your project in MPLAB.
- Click on *Begin uploading* and wait until the firmware is written on the board. After 5 seconds the board will execute your code

Upload your Firmware (Ubuntu)

To write your program on the board from a Ubuntu distro you must install an additional tool: https://github.com/thotypous/mikroe-uhb

Clone the *master* repository from the github link, the folder content will be the following:

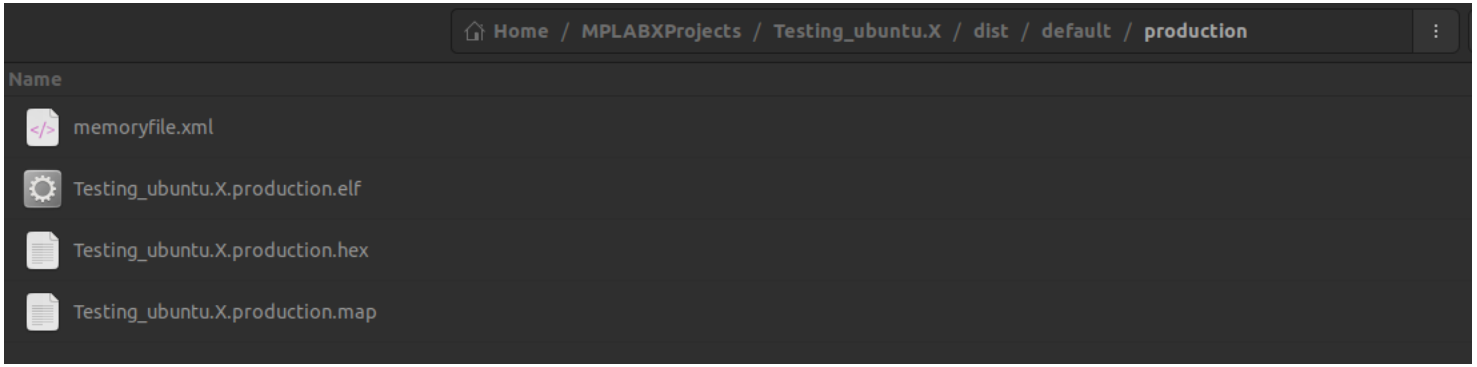


Open a terminal in the main folder and run *make* and then *sudo make install* commands. Note that you must have a Python installation on your system.

If you encounter issues in building the library, run *sudo pip install "setuptools<58.0.0"*

Upload your Firmware (Ubuntu)

To build your project on Ubuntu just follow the same steps shown in the Windows tutorial, then open a terminal in the folder containing the *.hex* file.



Upload your Firmware (Ubuntu)

Connect the board to the PC and run *sudo lsusb*: you need this step to know how the board is identified. The output should look like:

```
sudo lsusb
Bus 004 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 003 Device 002: ID 0c45:6a1b Microdia Integrated_Webcam_FHD
Bus 003 Device 004: ID 046d:c03e Logitech, Inc. Premium Optical Wheel Mouse (M-BT58)
Bus 003 Device 003: ID 8087:0033 Intel Corp.
Bus 003 Device 016: ID 2dbc:0001 Mikroelektronika USB HID Bootloader
Bus 003 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
```

You can notice that the Bootloader stored on the board is recognized as *Mikroelektronika USB HID Bootloader*. If that record does not appear in the listing, press the reset button on the board and re-run the lsusb command within 5 seconds from the button toggle.

Take note of the ID values in the listing, in this case *2dbc and 0001*.

Upload your Firmware (Ubuntu)

Now you can run the flashing command, remember that this is just an example and the *vendor id* and *product id* could be different on your machine.

```
francesco@francesco-Vostro-7620:~/MPLABXProjects/Testing_ubuntu.X/dist/default/production$ sudo mikroe-uhb -v --vend
or=0x2dbc --product=0x0001 Testing_ubuntu.X.production.hex
[sudo] password for francesco:
DEBUG:mikroeuhb.hid.linux:opening device vendor=2dbc, product=1
INFO:mikroeuhb.hid.linux:USB device 2dbc:0001 plugged
INFO:mikroeuhb.hid.linux:USB ID matches the expected one
DEBUG:mikroeuhb.device:send cmd: stx, cmd=INFO, addr=0x00000000, counter=0x0000
DEBUG:mikroeuhb.device:recv data: 3c010b0008000004080003000000c040080010500001306000040050007436c69636b6572203220666f
722064735049433333000000000000000000000549526cd
ERROR:mikroeuhb.bootinfo:Field 51 not recognized -- aborting parsing
McuType: 'DSPIC33'
EraseBlock: 0xc00
WriteBlock: 0x180
BootRev: 0x1300
BootStart: 0x54000
DevDsc: b'Clicker 2 for dsPIC3'
McuSize: 0x80400

DEBUG:mikroeuhb.device:send cmd: stx, cmd=BOOT, addr=0x00000000, counter=0x0000
DEBUG:mikroeuhb.device:recv cmd: stx, cmd=BOOT, addr=0x0008000b, counter=0x0400
DEBUG:mikroeuhb.device:send cmd: stx, cmd=SYNC, addr=0x00000000, counter=0x0000
DEBUG:mikroeuhb.device:recv cmd: stx, cmd=SYNC, addr=0x0008000b, counter=0x0400
DEBUG:mikroeuhb.devkit:reset code before fix: 000204000000
DEBUG:mikroeuhb.devkit:reset code after fix:  004004050000
DEBUG:mikroeuhb.devkit:transfer to device starting
```

At this point, the terminal will hang in loop until the on-board bootloader is detected, showing an *opening device* message. Just press again the reset button and the flashing process will begin. The output should be like the following screen.



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